

modules (5.7 m²) reduce the overall balance of system costs. The net effect is that the customer gets power cheaper.

The technology is expensive. How long will it take to make it affordable?

RT: Solar technology has already achieved grid parity in several areas today. For example, California's peak summer electricity rates hover above 35 cents/kWh in certain places, which is more expensive than unsubsidised solar power generated today at about 30 cents/kWh. In addition, solar power can be distributed across the grid to relieve pressure on critical substations or transmission lines without the time or expense of re-wiring new transmission or distribution facilities.

As a resource for energy generation, solar is incredibly abundant. Every day, the sun provides more energy than we need for the planet. What this means is that from a cost, availability and rapid-deployment perspective, solar power makes sense right now.

Additionally, thin-film technology dramatically reduces the amount of silicon needed to produce a module, thereby reducing costs. By applying our company's 40 years of experience in driving down the cost of integrated circuits, and more recently flat-panel displays, we're also bringing advances in deposition process, automation and factory efficiency to the thin-film production environment. All of these efforts further help to drive down the price of solar power generation.

What are your future plans with Applied Materials?

RA: We have just started production at this 40MW capacity line, which is the largest thin-film solar line in India. This single-junction line has demonstrated the highest production capacity to date for manufacturing the world's largest (2.2x2.6m²) solar modules.

Next, we are building the second plant in Greater Noida with AMAT. This is a 65MW tandem junction plant with the same large-footprint module. The full-size product will generate greater than half-a-kilowatt per panel. We are also in advance stages of discussions with AMAT on our 500MW thin-film plant in Chennai.

What trends do you foresee in the Indian solar PV space?

RA: The technological landscape of solar power is impressive: Crystalline silicon is the most mature technology and has the largest market share, but it is in short supply. Thin-film is a maturing technology and poised for high growth. Also, it has the potential to reduce costs significantly. Concentration PV is being rapidly developed for the market and it has significant cost-reduction potential as well. Finally, nanotechnology is in the R&D phase and it has tremendous cost-reduction potential.

As a trend we see more installations of solar PV for small, medium and large off-grid applications and large solar farms for on-grid applications. Our effort is to work diligently in all PV technologies and bring the solar dream to fruition to meet India's energy needs.

Introducing

CHIP-TALK Hobby Projects

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These projects have been selected from nearly a thousand circuits that have appeared in EYF magazine over the past decade or so. Chip-Talk gives unique opportunity to learn assembling of circuits to one and all, including experienced electronics people.

It is accompanied by a book which explains fundamentals of electronics along with workshop procedure to assemble the circuits. It offers a unique innovative method of learning for budding hobbyists and for all those who want to step into the enabling field of electronics.

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